

Mohammad Sadil Khan

Research Scientist in 3D Vision and Generative AI

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Research Scientist working on foundation models for spatial and physical intelligence, with focus on parametric CAD generation, multi-modal 3D reconstruction, world models, and scalable post-training through 3D reward modeling.

PUBLICATIONS

PEER-REVIEWED

CVPR 2024 ★ Highlight

CAD-SIGNet: CAD Language Inference from Point Clouds using Layer-wise Sketch Instance Guided Attention

Mohammad Sadil Khan, et al.

A SOTA architecture with a novel masked attention mechanism for generating 3D CAD models from point clouds.

📄 Paper | 🌐 Website

NeurIPS 2024 ★ Spotlight

Text2CAD: Generating Sequential CAD Models from Beginner-to-Expert Level Text Prompts

Mohammad Sadil Khan*, Sankalp Sinha*, et al. *equal contribution

First AI framework to generate 3D CAD designs from text descriptions.

📄 Paper | 🌐 Website | 💻 Code | 🗃 Dataset

CVPR 2025

MARVEL-40M+: Multi-Level Visual Elaboration for High-Fidelity Text-to-3D Content Creation

Sankalp Sinha*, Mohammad Sadil Khan*, et al. *equally contributing first authors

Largest 3D captioning dataset (40M) with domain-specific and multi-level annotation.

📄 arXiv | 🌐 Website | 🔍 Explorer | 🗃 Dataset

AAAI 2026

NURBGen: High-Fidelity Text-to-CAD Generation through LLM-Driven NURBS Modeling

Muhammad Usama*, Mohammad Sadil Khan*, et al. *equally contributing first authors.

First NURBS-based LLM assistant for text-to-CAD generation. Introduces the PartABC dataset with text captions.

📄 arXiv | 🌐 Website

UNDER REVIEW

Preprint

DreamCAD: Scaling Multi-Modal CAD Generation using Differentiable Parametric Surfaces

Mohammad Sadil Khan, et al.

State-of-the-art Bezier surface-based CAD representation. CADCap-1M: Largest CAD captioning dataset.

📄 arXiv | 🌐 Website

Preprint

BRepCLIP: Contrastive Multimodal Pretraining on BRep Primitives for CAD Understanding

Muhammad Usama, Didier Stricker, Mohammad Sadil Khan[†], Muhammad Zeshan Afzal[†] [†]Joint Supervision.

(1) Representation Learning for BRep models (2) New evaluation metric for text-to-CAD.

📄 arXiv | 🌐 Website

PATENT

Multimodal CAD Reconstruction Using Differentiable Parametric Surfaces

Filed 03/03/2026

Assignee: Huawei Technologies Co., Ltd. (Noah's Ark Lab, London, UK)

Inventors: Mohammad Sadil Khan (50%), Jiankang Deng (25%), Ismail Elezi (25%)

EXPERIENCE

Spectral Labs

Research Scientist - Full Time

London, UK

May 2026 –

- Designing foundation models for multi-modal CAD generation, including post-training and reward modeling strategies.

Noah's Ark Lab, Huawei

Research Intern - Internship

London, UK

May 2025 – Apr 2026

- Conducted research in 3D CAD Reconstruction under Dr Ismail Elezi (Principal Research Scientist) and Prof. Jiankang Deng (Asst. Prof. Imperial College London).

Augmented Vision Lab, DFKI

Research Assistant - Part Time

Kaiserslautern, Germany

Jun 2024 – Apr 2025

- Designed and implemented novel algorithms for AI-enhanced parametric CAD modelling for Luminous Project.

CVI2 Group, SNT, University of Luxembourg

Student Researcher - Full Time

Kirchberg, Luxembourg

Jan 2023 – Feb 2024

- Designed novel algorithms (CAD-SIGNet) for the Scan2CAD research project.
- Served as Technical Committee Member for the SHARP Workshop at ICCV 2023.

Creatis, INSA Lyon

Research Intern - Internship

Lyon, France

Feb 2022 – Jul 2022

- Developed a novel 3D medical image segmentation approach using point clouds.

EDUCATION

PhD in 3D Computer Science

RPTU Kaiserslautern & DFKI

Kaiserslautern, Germany

Feb 2024 – Present

Supervised by Prof. Didier Stricker. Focus: 3D Reconstruction, Generative AI, Parametric Shape Modeling.

M.Sc. in Machine Learning and Data Mining

University Jean Monnet & KU Leuven

France / Belgium

Sep 2020 – Jul 2022

Erasmus Scholarship for exchange semester

Masters (M1) in Data Science

Chennai Mathematical Institute

Siruseri, India

Aug 2019 – Jul 2020

B.Sc. in Mathematics

Ramakrishna Mission Residential College

Kolkata, India

Jul 2016 – May 2019

Rank 2nd

SKILLS

Architecture	VLM/LLM Inference, LoRA Fine-tuning, Transformers, Flow Matching, Diffusion Models
Libraries	PyTorch, PyTorch Lightning, Pytorch3D, vLLM, Blender Python, Kaolin, PythonOCC
Programming	Python, R, Julia, Java, Swift
Languages	Bengali (Native), Hindi (C2), English (C1), French (A2)

AWARDS & SCHOLARSHIPS

2021–2022 BRMI Regional Scholarship & Erasmus Scholarship, France

2020–2021 Charpak BCS Scholarship, French Embassy, India

ACADEMIC SERVICE

Peer Review : 15+ papers

CVPR, NeurIPS, ECCV, ICML, AAAI, SIGGRAPH Asia, ICPR

Mentorship : Student supervision

2025 – 2026

Workshop : Technical Committee Member

SHARP @ ICCV 2023